

Cyber Security Threats & Defense Guide

Short, Sweet, and Easily Understandable Threat Matrix with Red-Marked Terms

Quick Attack Categorization

Core Attack Domain	Included Cyber Exploits
Human-Based Attacks	Phishing, Social Engineering
Malware-Based Attacks	Ransomware, General Malware, Spyware / Keylogger
Web-Based Vulnerabilities	SQL Injection (SQLi), Cross-Site Scripting (XSS)
Network-Level Attacks	Man-in-the-Middle (MITM), DDoS Outages
Authentication Threat Vectors	Brute Force, Dictionary & Password Attacks
Advanced Dynamic Exploits	Zero-Day Flaws, Supply Chain Exploits, Rogue Botnets, Insider Threats

1. Phishing Attack

What is it? → Phishing ek simple **Social Engineering** attack chhe. Jema attacker fake email, fake login page, and fake websites direct user ne target karva mate use kare chhe. Target no objective sensitive data and private identification steal karvano hoy chhe.

Main Targets: Usernames, Private Passwords, Live OTP, Credit Card Info

Attack Sequence Flow Map:

Fake Bank Email Sent → User Clicks Deceptive Link → Fake Form Interface Appears → Credentials Stolen

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
Spear Phishing	Specific target configuration or targeted individual context
Whaling Threat	Targeting high-profile executives like company CEOs / MDs
Vishing	Voice call based scam interactions over telephones
Smishing	Malicious link integration shared via dynamic SMS messages

Prevention & Mitigation Action Controls:

- Always carefully double-check URLs before any login activity
- Enforce Multi-Factor Authentication (MFA) parameters sitewide
- Completely avoid opening dynamic unknown link paths

2. Ransomware Attack

What is it? → Ransomware user ni core files tone background ma background memory completely **Encrypt** (lock) kari dethi. Ane te primary systems open and unlock karva badle clear extortion money demand i.e. **Ransom Note** screen context upar display kare chhe.

Main Targets: Corporate Systems, Healthcare Enterprise Databases, Government Servers

Attack Sequence Flow Map:

Malware Enters Network → Files Get Cryptographically Locked → Ransom Note Displayed → Crypto Payment Demand

Real-World Operational Paradigm: WannaCry Ransomware (2017) global scale systems network collapse framework.

Prevention & Mitigation Action Controls:

- Maintain continuous offline standalone data backups regular model
- Keep software packages and patches completely updated
- Deploy corporate endpoint Antivirus shielding software

3. Malware Attack

What is it? → Malware generic text words mathi shorthand validation metadata **Malicious Software** chhe. Ahna custom hidden design patterns host computer system database data variables ne harm, corrupt or change karva mate specialized rules upar construct thay chhe.

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
Active Virus	Infects system files and reproduces instantly upon manual execution
Network Worm	Self-replicating rogue program that travels automatically across networks
Trojan Horse	Disguises as completely safe legitimate utilities but executes damage code in background
Secret Spyware	Monitors user behavior strings and coordinates asset logs silently

Real-World Operational Paradigm: Cracked patches software tools, key-generators, download setup packages inside systems installer files.

Prevention & Mitigation Action Controls:

- Download applications solely via vendor trusted certified stores
- Set system checks via continuous operational automated Antivirus engines
- Keep operational framework patch configuration sets updated periodically

4. DDoS Attack (Distributed Denial of Service)

What is it? → Multiple compromised computers through huge fake traffic streams generate kari target web servers validation capacity target limit memory full crash block execute configuration design model crash control. Server load control parameter system background data completely capacity down.

Main Targets: E-Commerce Portals, Online Multiplayer Gaming Systems, Banking Service Frontends

Attack Sequence Flow Map:

Rogue Botnet Array Triggered → Massive Fake Web Traffic Flood Ingestion → Server Core Resource Pool Exhaustion → Target Website Goes Completely Down

Prevention & Mitigation Action Controls:

- Utilize strong distributed Content Delivery Networks (CDN)
- Integrate customizable dynamic Web Application Firewalls (WAF)
- Enforce API structural layer Rate Limiting restrictions

5. SQL Injection (SQLi)

What is it? → Web fields input parameters parsing validation validation loopholes criteria check filter validation target backend structural execution parameters database engine framework direct manipulation process override database data unauthorized download system manipulation control schemas context layout access controls details check rules execute properties parameters criteria loop structure details logs metrics logic processing configuration updates layouts dynamic.

Main Targets: Application Input Authorization Modules, System Search Query Parameter Boxes

Real-World Operational Paradigm: Entering dynamic custom validation string characters like: ' OR '1'='1 inside login user text blocks.

Prevention & Mitigation Action Controls:

- Strictly mandate Parameterized Queries / Prepared Statements execution templates
- Ensure input validation structural verification filter checking checks everywhere

6. Password Attacks

What is it? → User system credentials passcodes target criteria tools through cracked algorithms guesses automated process layout logic configurations framework rules verify authorization bypass modules control.

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
Brute Force Method	Systematically testing every character set sequence variation array mathematically
Dictionary Schemes	Cycling queries through precompiled text glossaries of standard word lists
Credential Stuffing	Automated validation using previously leaked account lists across unrelated servers

Prevention & Mitigation Action Controls:

- Mandate multi-character strong passwords mixing numbers and symbols
- Incorporate systematic account lockout limits after limited failed trials
- Enforce Multi-Factor Authentication protection layers

7. Social Engineering

What is it? → Human behavior tracking manipulation mechanisms trust systems through critical internal passwords database information records share authorization context profiles.

Attack Sequence Flow Map:

Spoofed Tech Identity Contact Initiation → Manipulating Human Trust Parameters → Victim Yields Secret Corporate System Credentials

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
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Phishing Elements	Deceptive target generic email templates deployment vectors
Pretexting Scenarios	Inventing complex personalized situational cover stories to validate information extraction
Baiting Items	Dropping physically infected hardware devices like corporate storage USB keys in target range

Prevention & Mitigation Action Controls:

- Initiate frequent enterprise threat awareness training loops for office workers
- Establish cross-verification validation processes across identity interfaces

8. MITM (Man-in-the-Middle) Attack

What is it? → Attacker user connectivity data paths stream routing center positions inside secretly intercept reading modification transmission sequences dynamic mapping system profile monitoring log patterns context setups definitions layout parameter codes designs configurations data checks properties variables context models.

Attack Sequence Flow Map:

*User Data Outbound Pipeline Request → Attacker Capture Loop Interception → Manipulated Content
Relayed Destination Platform Server*

Real-World Operational Paradigm: Sniffing dynamic connection stream packets over unsecured open public wireless hotspots.

Prevention & Mitigation Action Controls:

- Enforce sitewide secure Transport Layer Security (HTTPS) connection structures
- Mandate operational routing architectures via virtual secure enterprise VPN tunnels
- Completely avoid connecting user configurations over open public Wi-Fi structures

9. XSS (Cross-Site Scripting)

What is it? → Vulnerable presentation code areas input fields script tags scripts inject code execute browser safe execution bypass parameters tracking details checks models logic setup contexts designs systems configurations profiles structure parameters metrics code.

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
Stored XSS	Malicious code is saved directly onto the server database tables permanent layout loading execution
Reflected XSS	Rogue dynamic string variables mirrored immediately back via direct link response values parsing parameters
DOM-Based XSS	Manipulating presentation execution contexts in client-side runtime scripts environment rules checking

Real-World Operational Paradigm: Injecting dynamic client side execution string triggers:
<script>alert("Hacked")</script>

Prevention & Mitigation Action Controls:

- Validate and sanitize application fields using intense Input Sanitization libraries
- Configure explicit contextual Output Encoding routines on template blocks

10. Zero-Day Attack

What is it? → Software vendor frameworks unpatched errors configurations definitions software bugs update code logic missing parameters design loopholes exploit execution before fix solutions patches available dynamic security structure signatures map context system monitoring tracking parameters.

Critical Danger Profile: No dynamic antivirus definitions or vendor software security fix blocks are physically ready at deployment time.

Prevention & Mitigation Action Controls:

- Establish dynamic heuristic behavioral pattern threat tracking configurations
- Formulate agile enterprise patch deployment engineering runbooks for critical infrastructure components

11. Botnet Attack

What is it? → Infected distributed internet connected devices framework grids zombie systems remote computer control interface master parameters access execute coordinated operations layouts rules structure setups.

Threat Use Cases: Massive DDoS Traffic Bombardment, Large Scale Email Spam Delivery, Large Volume Widespread Automated Credential Stuffing Logs

Prevention & Mitigation Action Controls:

- Enforce regular operational firmware version secure update routines
- Change system baseline admin configuration setup passwords across all local smart nodes

12. Insider Threat

What is it? → Organization internal authentic user accounts workers contractors personnel corporate entities administrative permission levels credentials misuse layout system context files leakage databases data modification systems compromise behaviors execution control track designs models parameters logic context structural paths definitions validation systems.

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
Malicious Actor	Intentionally exploiting access clearance permissions to steal intellectual secrets or disable corporate units

Careless Insider	Staff members executing unintentional system mistakes or falling prey to data collection setups blindly
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Prevention & Mitigation Action Controls:

- Rigidly structure permission matrix guidelines under the Principle of Least Privilege (PoLP)
- Deploy deep activity profile checking mechanisms paired with automated log collection systems

13. Wi-Fi Attacks

What is it? → Wireless radio communications protocol layer handshakes connection authentication procedures frame interception modifications credential cracking structures verification bypass rules designs systems configurations tracking profiles parameters metrics.

Attack Classifications & Variations:

Classification Model	Operational Concept Profile
WPA Pre-shared Key Cracking	Capturing live baseline security validation handshake blocks to run local offline cryptographic crunching checks
Evil Twin Traps	Configuring rogue identical network radio wireless names to capture target user connections automatically
Deauthentication Dropping	Flooding counterfeit protocol frames to forcefully drop targeted client machines from home routers

Prevention & Mitigation Action Controls:

- Upgrade legacy network operational standard options to the next-generation robust WPA3 encryption baseline
- Configure long and heavily nested network security access system keyphrases

14. Spyware / Keylogger

What is it? → System environment process tracing tracking modules invisible hidden execution background patterns record data strings file system key logs data logging configurations validation definitions profile details metrics parameters monitoring.

Operational Processing Engine: Directly tapping mechanical or onscreen keyboard input signals, saving character structures locally inside text files, and exfiltrating datasets to target domains.

Prevention & Mitigation Action Controls:

- Activate behavioral heuristic layer detection engines inside active Endpoint Security suites
- Block any peripheral installation requests derived from unauthorized third-party archive packages portal hubs

15. Supply Chain Attack

What is it? → Target organization direct network perimeter systems bypass instead trusted vendor software repositories compilation channels dependency updates distribution code integration pipelines system logic compromise execution configurations parameters models definitions profiles checks data properties vectors codes architecture.

Attack Sequence Flow Map:

Trusted Vendor Software Firm Hack → Malicious Backdoor Code Injected Into Certified Production Repositories → Client Endpoints Process Verified Signed Malicious Update Automatically → Downstream Target Enterprise Architecture Infestation Active

Real-World Operational Paradigm: SolarWinds security supply exploit chain breach instance (2020) triggering massive critical asset access tracking compromises globally.

Prevention & Mitigation Action Controls:

- Execute thorough structural binary verification checking logs on incoming runtime software distribution patches
- Establish consistent structural vendor posture verification metrics and security alignment auditing checks

Which Attack Mate Kai Tool Use Thay?

Attack	Kali Tool
Phishing	SEToolkit, Gophish
Ransomware Testing	Metasploit labs
Malware Analysis	YARA, Radare2
SQL Injection	SQLMap
XSS Testing	Burp Suite
Password Attack	Hydra, John the Ripper
Wi-Fi Attack	Aircrack-ng
MITM	Ettercap, Bettercap
Packet Sniffing	Wireshark
Reconnaissance	Nmap
Exploitation	Metasploit
Web Testing	Burp Suite
Vulnerability Scanning	Nikto, OpenVAS

Example Workflow in Kali Linux

Reconnaissance

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Nmap Scan

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Find Open Ports

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Vulnerability Testing

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Exploitation (Lab)

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Reporting